

None of these complex functions are controlled by a single neural structure. Rather, they are emergent properties of the interactions among widespread regions in the nervous system. There is ample evidence that coordinated interactions between the cerebral cortex and the thalamus allow us to link opportunities for action (such as a steering wheel and pedals) with goals and plans (driving a car to get to the store). However, these two major neural systems don't work alone. For instance, both structures are strongly [connected](#) to the cerebellum, a massive structure at the base of the brain. The cerebellum [creates](#) a high-resolution copy of the actions that we take in a given context, and then [learns](#) to recombine features of the situation and our planned action into the best response for what to do next in that situation. In contrast, inhibitory circuits within the [thalamus](#) and [cerebral cortex](#) are better suited for ensuring that different sequences (eg, steering the car and listening to music) don't blend into one another.

The ability to control the balance between automatic and deliberate processing is [well suited](#) to the neuromodulatory arousal system. This system is made up of nuclei (collections of nerve cells) that project widely throughout the brain, wherein they release neurochemicals that [change](#) the way that different regions interact. Based on their widespread connectivity, these structures provide modulatory control of all of the other processes. For instance, the neurochemical noradrenaline controls key features of arousal, whereas dopamine is more [important](#) for motivation.